

Direction of Arrival Tracking using Orthogonal Dipoles

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Abstract—In this paper, we propose a novel approach for tracking the direction of arrival (DOA) using a collocated microphone array equipped with double dipole microphones. Our method consists of two main steps. First, we employ a DOA estimation procedure based on a histogram-based wrapped Gaussian mixture model (Hist-WGMM). Second, we introduce a new Kalman filter based on the wrapped Gaussian distribution. The DOA estimation procedure is designed for real-time deployment and exhibits robustness against reverberation and noise. Additionally, our proposed Kalman filter mitigates the issue of oscillations around 0° , a common problem faced by conventional Kalman filters. Through numerical simulations, we validate the effectiveness of our entire tracking method, demonstrating its ability to accurately estimate and track DOA in a fast and precise manner.

Index Terms—DOA estimation, DOA tracking, orthogonal dipoles, Kalman filter, mixture model

I. INTRODUCTION

Differential microphone arrays (DMAs) are extensively employed in speech processing. In [1], [2], an adaptive DMA was devised to mitigate the interference of undesired speeches on the desired speech signal. Furthermore, DMAs are utilized to estimate the power spectral density (PSD) of ambient noise and improve speech quality by subtracting noisy spectra [3], [4]. Theoretical investigations have been conducted on the design and implementation of linear DMAs [5]. Over the past decade, a variant of DMA, the circular differential microphone array (CDMA), has garnered increased attention in audio processing. A general approach to designing frequency-invariant beam patterns such as dipoles and cardioids has been developed using uniform CDMA in [6]. Specifically, a uniform CDMA comprising 4 omni microphones distributed evenly on a circle with a radius of 2cm has been employed to create two orthogonal dipole beam patterns [7], [8]. Based on these two dipoles, a DOA estimation method utilizing error reduction functions, denoted as DOA-IOD, has been proposed to locate the speaker [9].

However, the aforementioned uniform CDMA with a diameter of 4cm still requires a relatively large space for some edge devices like wearable sensors. Moreover, the relative position between sound sources and microphones usually changes in practice. Additionally, the aforementioned work [9] does not address the issues of tracking DOA estimates. On the other hand, benefiting from the development of microelectromechanical systems (MEMS), directional microphones can form frequency-independent dipole beam patterns [10]. This implies that the two orthogonal dipoles can be achieved by placing these collocated directional MEMS acoustic sensors together. However, there is a lack of investigations into how to estimate and track DOAs using such a collocated array.

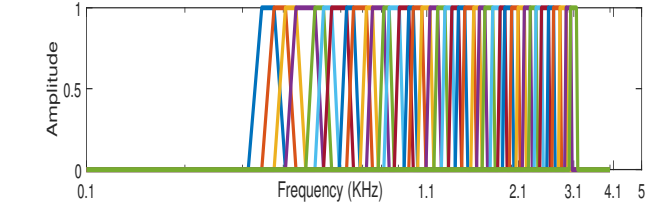


Fig. 1: The 40 bandpass filters

In this paper, by employing two collocated dipole microphones, we design a more compact microphone array, requiring less space than the uniform DMA does. Moreover, we propose corresponding methods for DOA estimation and tracking, utilizing a histogram-based wrapped Gaussian mixture model (Hist-WGMM) and a wrapped Kalman filter (WKF), respectively, under the presence of reverberation and white noise. Numerical simulations are conducted, with results demonstrating that the proposed DOA estimation method outperforms state-of-the-art methods like DOA-IOD, and the proposed WKF provides more precise tracking results than the conventional Kalman filter (CKF).

II. PROPOSED DOA ESTIMATION METHOD

A. Signal model

Two collocated dipole microphones can form two collocated orthogonal dipoles (X-Dipole and Y-Dipole). The signals obtained from X-Dipole $x(n)$ and that from Y-Dipole $y(n)$ are, respectively, given by:

$$x(n) = \cos\theta \cdot s(n) + n_x(n) \quad (1)$$

$$y(n) = \sin\theta \cdot s(n) + n_y(n) \quad (2)$$

where $s(n)$ denotes the speech signal coming from angle θ ; $n_x(n)$ and $n_y(n)$ represent the white noise signals in the sensors forming X-Dipole and Y-Dipole, respectively.

By applying short time Fourier transform (STFT) to both $x(n)$ and $y(n)$, we obtain $X(t, f)$ and $Y(t, f)$, where t and f denotes time and frequency index respectively. As illustrated in

Fig.1, by utilizing 40 rectangular bandpass filters to divided the whole frequency band and employing the energy ratio method [11], we can obtain the DOA estimate of the p th subband signal as

$$\theta(t, p) = \begin{cases} \tan^{-1}(Q), & \text{xcor}(x(n), y(n)) > 0 \\ \pi - \tan^{-1}(Q), & \text{otherwise} \end{cases} \quad (3)$$

with $Q = \sqrt{\frac{E_Y(t, p)}{E_X(t, p)}}$, where $E_Y(t, p)$ and $E_X(t, p)$ denote the energy terms, as calculated by $|Y(t, f)|^2 \cdot B(f, p)$ and $|X(t, f)|^2 \cdot B(f, p)$, respectively; xcor represents the cross correlation function; $B(f, p)$ denotes the p th bandpass filter implemented in frequency domain. The complete procedure to obtain these bandpass filters is outlined in Alg.1, where Fs is the sampling rate; N_{fft} is the number of points for fast Fourier transform (FFT); N_{filter} is the number of filters; f_{min} and f_{max} are the minimal and maximal frequencies, respectively. Note that Eq.(3) introduces the phase ambiguity

Algorithm 1 Rectangular bandpass filters [11]

Input: $Fs, N_{fft}, f_{min}, f_{max}, N_{filter}$

Procedure:

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 $m_{min} \leftarrow 2595 \log_{10}(1 + \frac{f_{min}}{700})$ 
 $m_{max} \leftarrow 2595 \log_{10}(1 + \frac{f_{max}}{700})$ 
 $m \leftarrow \text{linspace}(m_{min}, m_{max}, N_{filter} + 1)$ 
 $f \leftarrow 700(10^{\frac{m}{2595}} - 1)$ 
 $b \leftarrow \text{floor}(\frac{f}{Fs} N_{fft}) + 1$ 
 $f_{start} \leftarrow b(1 : \text{end} - 1)$ 
 $f_{stop} \leftarrow b(2 : \text{end})$ 
 $B \leftarrow \text{zeros}(N_{fft}, N_{filter})$ 
 $p \leftarrow 1$ 
while  $p \leq N_{filter}$  do
     $B(f_{start}(p) : f_{stop}(p), p) \leftarrow 1$ 
     $p \leftarrow p + 1$ 
end while
return  $B$ 

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problem between 0 and π , implying that the $\theta(t, p)$ can be either 0 or π if Q equals 0. Furthermore, due to the presence of noise and reverberation, the raw DOA estimates calculated by Eq.(3) are highly noisy. To mitigate this issue, we introduce a threshold $E_T(p)$, represented as a p -dimensional vector obtained through microphone calibration. Here, $E_{am}(p)$ is computed by averaging the maximum values of $E_X(t, p)$ and $E_Y(t, p)$ over t , following the recording of a speech signal. We set $E_T(p)$ to $k \cdot E_{am}(p)$, where $k \in [0, 1]$, to filter out unreliable DOA values $\theta(t, p)$. Specifically, $\theta(t, p)$ is discarded only if both $E_X(t, p)$ and $E_Y(t, p)$ are less than $E_T(p)$. Subsequently, the remaining raw DOA values obtained by Eq.(3) are utilized to infer the ground truth DOA through a clustering method based on a probabilistic mixture model.

B. Histogram-based mixture model

The wrapped Gaussian mixture model (WGMM) [12] is widely used for clustering angular (circular) values. Since the DOA estimate possesses angular properties, the WGMM

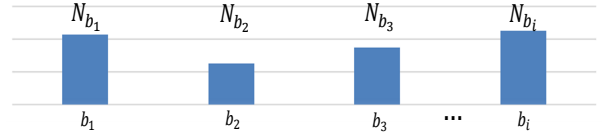


Fig. 2: The histogram representation

serves as a robust DOA estimation method [13]. However, the solving procedure for WGMM relies on the expectation-maximization (EM) algorithm [14], an iterative technique comprising E-step and M-step in each iteration to compute maximum likelihood estimates. From a time complexity perspective, the EM algorithm can be bounded by $O(I_{max} \cdot N_{sample})$, where I_{max} and N_{sample} denote the maximal number of iterations and the number of samples, respectively. In this section, we employ a histogram to represent the distribution of samples and propose the Hist-WGMM, whose time complexity is bounded by $O(I_{max} \cdot N_{bin})$, where N_{bin} is the number of bins in a histogram.

1) *Histogram representation*: We utilize a histogram to represent the distribution of DOA samples, as illustrated in Fig.2, where b_i denotes the i th bin in the histogram and N_{b_i} denotes the number of samples in b_i . In other words, there are N_{b_i} samples, whose values are identical to θ_i , in the i th bin b_i . Additionally, $\sum_{i=1}^{N_{bin}} N_{b_i} = N_{sample}$.

2) *Wrapped Gaussian distribution (WGD)*: Given a DOA θ^i , the probability density function (PDF) of the WGD is defined as:

$$f(\theta^i; \mu, \sigma^2, T) \propto \sum_{w \in Z} \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(\theta^i - w \cdot T - \mu)^2}{2\sigma^2}} \quad (4)$$

where μ and σ^2 denote the mean and variance, respectively; T represents the period, which is fixed as π in this paper due to the phase ambiguity introduced in Eq.(3).

In the case of multiple sound sources present concurrently, we model the distribution of DOA samples from the k th source as a k th WGD. Then, assuming that there are K sound sources in total, the probability of θ_i based on the total probability rule is given by:

$$P(\theta^i; \beta, T) = \sum_{k=1}^K f(\theta^i; \mu_k, \sigma_k^2, T) \cdot \alpha_k \quad (5)$$

where β represents the parameter set $\{\alpha_k, \mu_k, \sigma_k^2\}$; μ_k and σ_k are the mean and variance of the k th WGD, respectively; α_k is the weight of the k th WGD. Now the DOA estimation problem can be reconfigured as an optimization problem to find the optimal parameter set β . More specifically, the mean μ_k can be regarded as a good DOA estimate of the k th sound source.

3) *EM algorithm*: Now we adapt the EM algorithm [14] to find the optimal β in a histogram-based approach. The cost

function of maximum log-likelihood estimation is given by:

$$L(\beta, \theta) = \sum_{i=1}^{N_{bin}} N_{b_i} \cdot \sum_{k=1}^K R_k^i \cdot [\log(\alpha_k) + \log(f(\theta^i; \mu_k, \sigma_k^2, T))] \quad (6)$$

where N_{bin} denotes the total number of bins in the histogram; and R_k^i can be calculated by $r(\theta^i, \beta, T) = \frac{f(\theta^i; \mu_k, \sigma_k^2, T) \cdot \alpha_k}{P(\theta^i)}$, which represents the posterior probability in Bayesian statistics. More specifically, μ_k can be regarded as a good DOA estimate for the k th sound source. To find the optimal β , which includes $\alpha_k, \mu_k, \sigma_k^2$, we first obtain two equations by setting the derivatives $\frac{dL}{d\mu_k}$ and $\frac{dL}{d\sigma_k^2}$ to 0, respectively. Then, by solving these equations, we can derive the expressions for μ_k and σ_k^2 as shown in Eq.(7) and Eq.(8), respectively.

$$\mu_k = \frac{\sum_{i=1}^{N_{bin}} D_i \sum_{w \in Z} e^{\frac{-(\theta^i - w \cdot T - \mu_k)^2}{2\sigma_k^2}} (\theta^i - w \cdot T)}{\sum_{i=1}^{N_{bin}} D_i \sum_{w \in Z} e^{\frac{-(\theta^i - w \cdot T - \mu_k)^2}{2\sigma_k^2}}} \quad (7)$$

$$\sigma_k^2 = \frac{\sum_{i=1}^{N_{bin}} D_i \sum_{w \in Z} e^{\frac{-(\theta^i - w \cdot T - \mu_k)^2}{2\sigma_k^2}} (\theta^i - w \cdot T - \mu_k)^2}{\sum_{i=1}^{N_{bin}} D_i \sum_{w \in Z} e^{\frac{-(\theta^i - w \cdot T - \mu_k)^2}{2\sigma_k^2}}} \quad (8)$$

where $D_i = \frac{N_{b_i}}{P(\theta^i; \beta, T)}$. The procedure to obtain the expression of α_k involves the Lagrange multiplier method [15] under $\sum_{i=1}^K \alpha_k = 1$, which is omitted due to limited pages. Here, we directly provide the expression below:

$$\alpha_k = \frac{\sum_{i=1}^{N_{bin}} N_{b_i} \cdot r(\theta^i, \beta, T)}{\sum_{i=1}^{N_{bin}} N_{b_i}}. \quad (9)$$

Now, the proposed iterative procedure using the EM algorithm is listed in Alg.2, where I_{max} is the maximal number of iterations; n denotes the iteration step; m is the convergence flag; ϵ is the convergence criterion. There are two major steps during each iteration: E-Step and M-Step. In the E-Step, the probabilities are firstly computed by Eq.(5). Then, the parameter set β is updated by Eq.(7), Eq.(8) and Eq.(9) in the M-Step. Moreover, the iterations in both E-step and M-step only need to scan all bins in the histogram. The overall time complexity of Alg.2 is bounded by $O(I_{max} \cdot N_{bin})$, which is suitable for real-time applications.

III. PROPOSED DOA TRACKING METHOD

In the context of Bayesian filters, the DOA estimates obtained in section II can be regarded as observations. This implies that the tracking procedures like the Kalman filter [16] can be applied to these observations to achieve better tracking results. However, due to the phase ambiguity introduced in Eq.(3), the assumption based on Gaussian distributions of conventional Kalman filter does not hold any more. Instead, the wrapped Kalman filter (WKF) [17] is the appropriate choice for tracking DOAs. In this section, we extend the theory of WKF, which can solve phase ambiguity between 0 and 2π during tracking, and generalize it to be feasible in phase ambiguity between 0 and T . We fix T as π in this paper.

Algorithm 2 Histogram-based EM algorithm to estimate β

Input: $\epsilon, \theta, K, \beta^0$

Procedure:

$\beta^n \leftarrow \beta^0$

$m \leftarrow \text{false}$

while $n \leq I_{max}$ and $\neg m$ **do**

$L_n \leftarrow L(\beta^n, \theta)$

$P(\theta^i; \beta, T) \leftarrow P(\theta^i; \beta^n, T)$ ▷ E-Step

$r(\theta^i, \beta, T) \leftarrow r(\theta^i; \beta^n, T)$ ▷ E-Step

$\beta^{n+1} \leftarrow \{\alpha_1, \dots, \alpha_K, \mu_1, \dots, \mu_K, \sigma_1^2, \dots, \sigma_K^2\}$ ▷ M-Step

$L_{n+1} \leftarrow L(\beta^{n+1}, \theta)$

if $|L_n - L_{n+1}| < \epsilon$ **then**

$m \leftarrow \text{true}$

end if

$n \leftarrow n + 1$

end while

A. Wrapped Kalman Dynamic (WKD)

The hidden state variable $x_t = [\theta_t \ \dot{\theta}_t]^T$ at time step t follows the WKD, which is expressed as:

$$x_t = A \cdot x_{t-1} + g_{t-1} \quad (10)$$

$$y_t = H \cdot x_t + m_t \quad (11)$$

where A is the transition matrix $\begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}$; Δt is the time interval between each observation; g_{t-1} is the process noise and $g_{t-1} \sim N(0, G)$; G is the covariance matrix of g_{t-1} ; H is the measurement matrix $H = [1 \ 0]$; m_t is the measurement noise and $m_t \sim N(0, \sigma_m^2)$; y_t denotes the observation value in the measurement step. Note that y_t is identical to the estimated DOA by the proposed Hist-WGMM at time step t .

B. Wrapped Kalman Filter (WKF)

Based on the WKD, the procedure of WKF, which includes prediction and correction steps, is listed in Alg.3, where P_t

Algorithm 3 Wrapped Kalman filter

Prediction Step:

$$x_t = A \cdot x_{t-1}$$

$$P_t = A \cdot P_{t-1} \cdot A^T + G$$

$$\theta_t = \psi_T(\theta_t), \text{ where } \theta_t \text{ in } x_t = [\theta_t \ \dot{\theta}_t]^T$$

Correction Step:

$$K_t = P_t \cdot H^T \cdot S^{-1}, \text{ where } S = H \cdot P_t \cdot H^T + \sigma_m^2$$

$$V_t = \frac{\sum_{w \in Z} (y_t - w \cdot T - H \cdot x_t) \cdot e^{\frac{-(y_t - w \cdot T - H \cdot x_t)^2}{2\sigma_m^2}}}{\sum_{w \in Z} e^{\frac{-(y_t - w \cdot T - H \cdot x_t)^2}{2\sigma_m^2}}}$$

$$x_t = x_t + K_t \cdot V_t$$

$$\theta_t = \psi_T(\theta_t), \text{ where } \theta_t \text{ in } x_t = [\theta_t \ \dot{\theta}_t]^T$$

$$P_t = (I - K_t \cdot H) \cdot P_t$$

denotes the state covariance matrix; I is the 2D identity matrix; K_t denotes the Kalman gain; V_t is the error term; $\psi_T(\theta_t) = \text{mod}(\theta_t + 0.5 \cdot T, T) - 0.5 \cdot T$ is an unwrapping function mapping θ_t into the range from $-\frac{T}{2}$ to $\frac{T}{2}$.

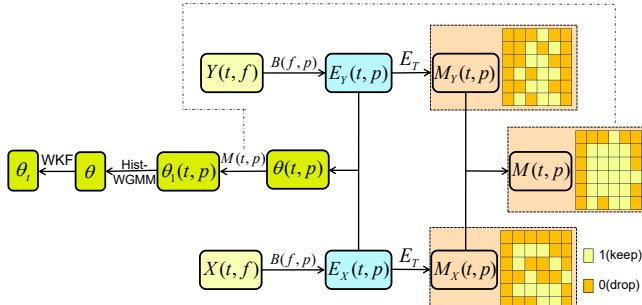


Fig. 3: The entire proposed DOA tracking method

TABLE I: The averaged running time (ms)

number of samples	WGMM	Hist-WGMM
500	12.85	2.01
1000	24.48	2.12
5000	146.30	5.76
10000	277.10	7.47
50000	1245.10	26.10

IV. THE ENTIRE PROCEDURE

We've consolidated all components of the proposed DOA tracking method into a diagram, depicted in Fig.3. Firstly, the bandpass filter $B(f, p)$ is applied to the time-frequency representations, $X(t, f)$ and $Y(t, f)$, resulting in $E_X(t, p)$ and $E_Y(t, p)$ respectively. Next, the threshold value $E_T(p)$, derived from microphone calibration, is applied to $E_X(t, p)$ and $E_Y(t, p)$ to obtain the masks $M_X(t, p)$ and $M_Y(t, p)$, respectively. These masks are combined using the logical OR operator to produce the final mask $M(t, p)$, i.e., $M(t, p) = M_X(t, p) \vee M_Y(t, p)$. This mask, $M(t, p)$, is then applied to $\theta(t, p)$, calculated from $E_X(t, p)$ and $E_Y(t, p)$ based on Eq.(3), to derive $\theta_1(t, p)$, serving as the input for the Hist-WGMM. The DOA estimate, θ , from the Hist-WGMM is treated as an observation value by the WKF, which yields the tracking result, θ_t .

V. EXPERIMENTS

A. Performance of the proposed Hist-WGMM

1) *time complexity*: We first construct 5 synthesized DOA datasets containing 500, 1000, 5000, 10000 and 50000 samples, respectively. In each dataset, DOA samples are synthesized using the parameters: $K = 2$, $\alpha_1 = 0.4$, $\alpha_2 = 0.6$, $\mu_1 = 135^\circ$, $\mu_2 = 0^\circ$, $\sigma_1 = \sigma_2 = 5^\circ$. We run the WGMM and proposed Hist-WGMM 200 times, respectively, on each dataset and record the corresponding running time. The averaged running time is listed in Table.I. The resolution and maximal number of iterations in the Hist-WGMM are set to 1° and 200, respectively. As shown in Table.I, compared with the WGMM, the proposed Hist-WGMM exhibits stable running time as the size of the synthesized dataset grows. This property demonstrate that the Hist-WGMM is suitable for use in real-time applications.

2) *performance*: Numerical simulations are conducted by simulating Room Impulse Responses (RIRs) using the image method [18], [19]. We selected the clean speech segments from the TIMIT [20] dataset as the signals of sound source. These clean speech signals were resampled to a frequency of 8 kHz. Gaussian noise was added to obtain different signal-to-noise ratios (SNRs). We obtained the time-frequency representation $X(t, f)$ and $Y(t, f)$ by using STFT with a 256-point Fast Fourier Transform (FFT), a Hamming window, and a frame shift of 128 points to achieve 50% frame-wise overlapping. The root mean square error (RMSE) is used as the metric to evaluate the performance of DOA estimation. To gather enough raw DOA samples, at the t th frame, we use the current $\theta(t, p)$ and that of the previous 24 frames to make a prediction using Hist-WGMM, where the resolution between bins in the histogram is 1° .

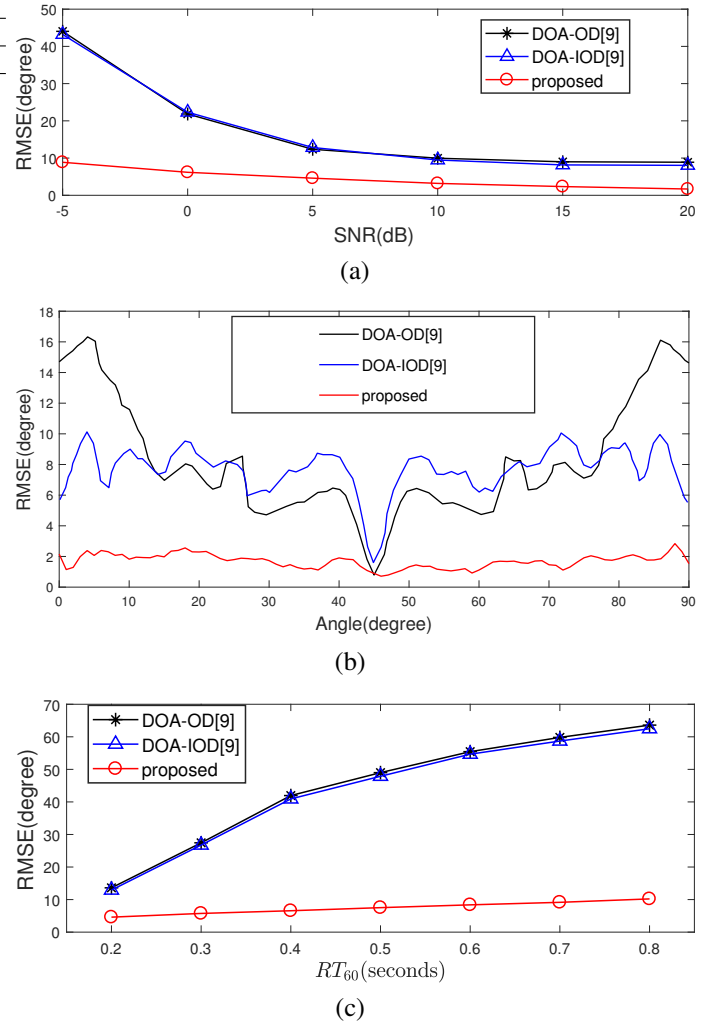


Fig. 4: The frame-wise performance comparison: (a)The averaged RMSEs under different SNRs (b)The RMSEs under different DOAs (c)The averaged RMSEs under different RT_{60} s

Firstly, we evaluate the performance under different SNRs, ranging from -5dB to 20dB. The speech comes from 0° to 90° with 1° resolution. We averaged the RMSEs of the DOAs per

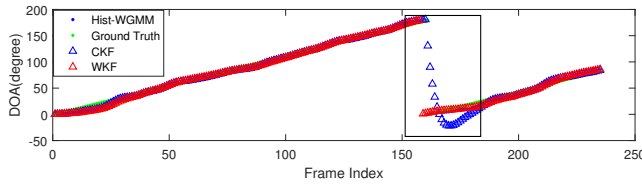


Fig. 5: Tracking performance comparison

SNR condition and plotted the results in Fig.4(a). Clearly, the Hist-WGMM outperforms the DOA-OD [9] and DOA-IOD [9] regardless of changes in SNR.

Secondly, we varied the speech incoming angle from 0° to 90° and calculated the frame-wise RMSEs under SNR=20dB. The resulted curves are shown in Fig.4(b), where the Hist-WGMM achieves better estimation regardless of the speech incoming angle and does not suffer dramatic performance loss around 0° and 90° as the DOA-OD [9] does.

Thirdly, to test the robustness of our proposed method to different reverberation intensities, we varied the RT_{60} from 0.2s to 0.8s under SNR=5dB and conducted simulations. The RMSE value averaged over 0° to 90° is also used as the performance metric. We plot the results in Fig.4(c). It is clear that the DOA-OD and DOA-IOD suffer much larger errors as the reverberation intensity becomes stronger. However, the Hist-WGMM demonstrates an extremely stable performance even in the strongest reverberation condition. This result shows our method is robust to reverberation in the presence of white noise. The robustness of our method to reverberation is attributed to the parameter σ_k^2 in Eq.(5), which can serve as an indicator of reverberation. Stronger reverberation yields a larger σ_k^2 , but μ_k , as the DOA estimate, is not influenced to the same extent.

B. Performance of proposed DOA tracking method

A virtual acoustic room with dimensions of $6\text{m} \times 5\text{m} \times 4\text{m}$ is constructed using the image method [18]. The collocated dipoles and the speaker are placed at (2m,1.5m,1.7m) and (3.25m,1.5m,1.7m), respectively. And the RT_{60} and SNR are set to 0.2s and 15dB, respectively, while the sound speed is fixed as 340m/s. The speaker is assumed to walk along a circle with (2m,1.5m,1.7m) as the center point and 1.25m as the radius. At each time frame, we regard the DOA estimate obtained by the Hist-WGMM as the observations, while using WKF and CKF to track the DOAs, respectively. The tracking result of this simulation is depicted in Fig.5. From the 0th to 150th time frame, the WKF and CKF obtain the same tracking performance. However, the phase ambiguity problem occurs around the 160th time frame, where the CKF produces large oscillations while the proposed WKF gives a stable and precise result. It is evident that the proposed WKF provides a better tracking result in the presence of phase ambiguity.

VI. CONCLUSIONS & FUTURE WORK

We propose a DOA tracking method for double collocated dipole microphones. This method includes two steps: DOA

estimation and the WKF. To estimate DOA, the Hist-WGMM is proposed to provide fast and robust estimation. We use the DOA estimated by Hist-WGMM as the input for the proposed WKF. The simulated results demonstrate that the entire tracking approach can provide accurate results. Additionally, the proposed Hist-WGMM supports locating multiple sound sources simultaneously, but the WKF does not support tracking multiple DOAs simultaneously. Tracking multiple DOAs using orthogonal dipoles is listed as a focus area for our future work.

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